

Control of a Nuclear Reactor Under Uncertainty

An MDP-Based Approach

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1. Introduction

Nuclear power plants generate electricity by using a controlled fission reaction to produce heat that drives a turbine. The amount of power output is regulated by control rods inserted into the reactor core. The more rods that go in, the lower the output.

The challenge is that control rod actions do not always produce the exact intended result. Trying to raise power by one level might raise it by two instead, or do nothing at all. This unpredictability makes the reactor hard to control, especially when it must continuously track a changing energy demand.

This system uses a Markov Decision Process (MDP) to handle that uncertainty. An MDP finds the best action to take in every possible situation, even when outcomes are random.

2. Tasks

2.1 MDP Modeling

2.1.1 Formal Definition

An MDP is defined by four components: (S, A, T, R)

- **S**: set of all possible states
- **A**: set of possible actions
- **T**: transition probability function
- **R**: reward function

The goal is to find an optimal policy $\pi^* : S \rightarrow A$ that tells the control system which action to take at each power level to minimize the gap between actual reactor output and current demand d_t . The discount factor was set to $\gamma = 0.9$ in all experiments (meaning that the system cares about the future, this matters because a bad action can cause problems several steps later.)

The reactor power is divided into 100 discrete levels numbered 0 to 99. Level 0 covers 0% to 1% power, level 1 covers 1% to 2%, and so on.

$$|S| = 100$$

Three control actions are available:

- **d (decrease)**: lower power by one level
- **m (maintain)**: hold power constant
- **i (increase)**: raise power by one level

$$|A| = 3$$

2.1.2 Transition Matrix P and Cost of Matrix C

Transition Matrix

Shape: $3 \times 100 \times 100 = 30,000$ entries.

Each entry $P[a, s, s']$ is the probability of going from state s to state s' when action a is taken. Each action has three possible outcomes with probabilities defined in the reactor's configuration file.

Table 1: Transition outcomes for each action

Action	Outcome	Probability	Intended?
decrease (d)	$s - 2$	$p_d[0]$	No
decrease (d)	$s - 1$	$p_d[1]$	Yes
decrease (d)	s	$p_d[2]$	No
maintain (m)	$s - 1$	$p_m[0]$	No
maintain (m)	s	$p_m[1]$	Yes
maintain (m)	$s + 1$	$p_m[2]$	No
increase (i)	s	$p_i[0]$	No
increase (i)	$s + 1$	$p_i[1]$	Yes
increase (i)	$s + 2$	$p_i[2]$	No

Boundary rule: Transitions below state 0 are clamped to 0, and transitions above state 99 are clamped to 99. Probabilities are accumulated at the boundary state rather than lost, so every row of P still sums to 1.0.

Cost matrix

Shape: $100 \times 3 = 300$ entries.

For each (state, action) pair, the reward is the expected negative distance between the demand and the next state's normalized power. Transitions that actively move the reactor away from the demand are penalized with a $\times 2$ multiplier on their distance:

$$\text{distance}(s', d_t) = \left| d_t - \frac{s'}{100} \right| \quad (1)$$

$$\text{cost}(s, s') = \begin{cases} 2 \times \text{distance} & \text{if moving away from } d_t \\ \text{distance} & \text{otherwise} \end{cases} \quad (2)$$

$$C(s, a) = \sum_i p_a[i] \times (-\text{cost}(s, s'_i)) \quad (3)$$

A transition is considered “moving away” when the reactor went up but the next power is already above demand, or went down but the next power is already below demand. If the state did not actually change, no penalty is applied.

C is recalculated at every time step because d_t changes. P is computed only once because the reactor physics never change.

2.1.3 Graph-Based Visual of MDP

The MDP can be visualized as a chain of 100 nodes, one for each state from 0 to 99. For any interior state s , each action creates directed edges to up to three neighboring states

based on the transition probabilities. At boundary states 0 and 99, edges that would go out of range fold back as self-loops, accumulating probability there.

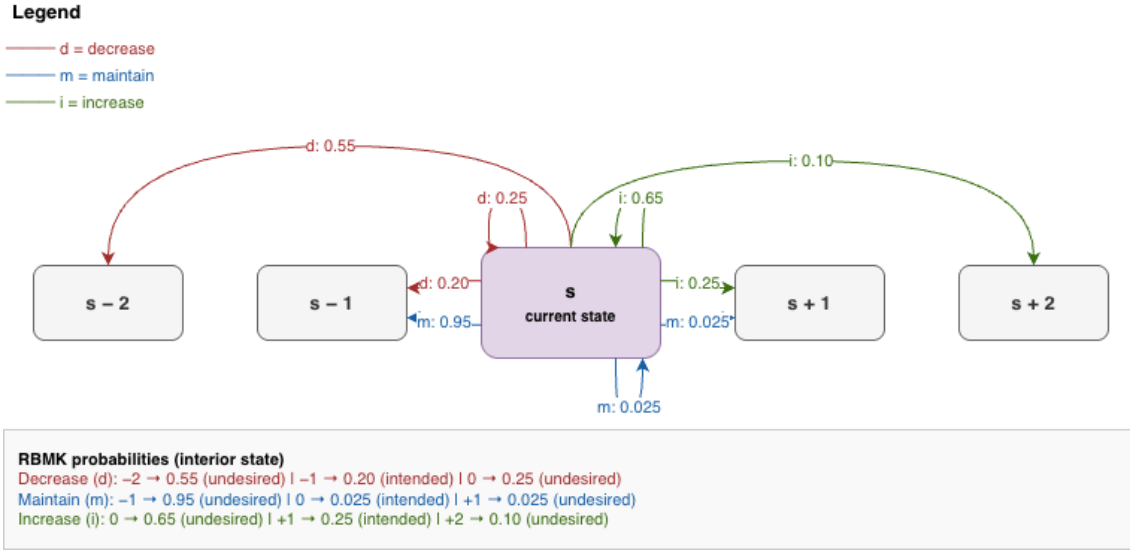


Figure 1: This represents the MDP topology for an interior reactor state. The diagram shows all possible state transitions produced by three actions: decrease, maintain, and increase, together with their associated probabilities.

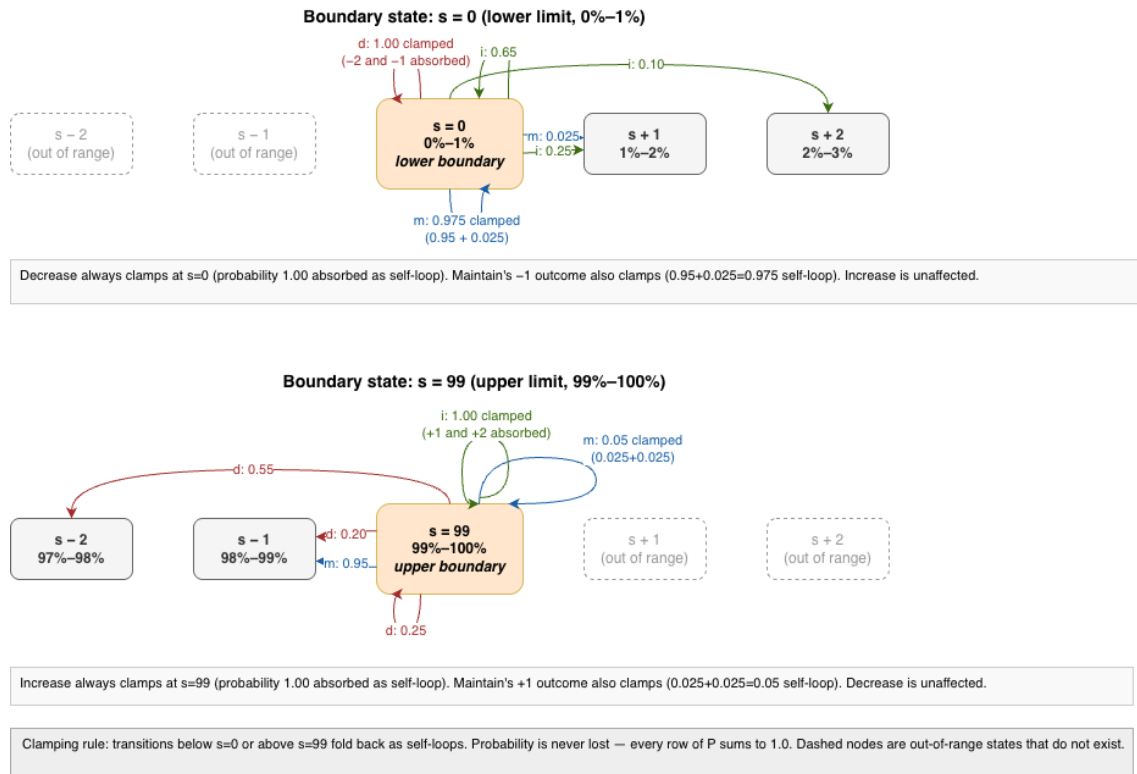


Figure 2: This represents the MDP topology at the reactor boundaries ($s=0$ and $s=99$). Invalid transitions outside the allowed state range are clamped back to the boundary state as self-loops to preserve valid probability distributions.

2.2 MDP Generation by Control Iteration

generate_P: Builds the $3 \times 100 \times 100$ transition matrix from the reactor's probability tables. Computed once per run since the reactor hardware does not change.

Suppose the reactor is currently at state 50 and the increase action has probabilities $[0.65, 0.25, 0.10]$. The transition matrix stores:

$$P[i, 50, 50] = 0.65$$

$$P[i, 50, 51] = 0.25$$

$$P[i, 50, 52] = 0.10$$

This means the reactor has a 65% probability of staying at the same level, a 25% probability of increasing by one level, and a 10% probability of overshooting by two levels.

generate_R: Builds the 100×3 reward matrix for the current demand value d_t . For each (state, action) pair it computes the expected negative distance across all three possible outcomes, doubling the distance for any outcome that moves the reactor away from the current demand. Recomputed at every time step. For a 512-step simulation, **generate_P** runs once and **generate_R** runs 512 times. P never changes between steps but R does, since the demand target is different at each step. Similarly, for any simulation where there are N demand samples, **generate_P** runs once while **generate_R** runs N times.

control_iteration: Takes the precomputed P , the current demand, and the current reactor state. Calls **generate_R** to build the reward matrix, then runs Value Iteration to find the optimal action for every state, and returns only the action for the current state.

2.3 Control Loop Integration

control_loop: Runs the complete simulation on all demand samples.

1. Initialize the reactor state from the first demand sample. Starting the reactor near the first demand value avoids unrealistic startup errors (that would occur if it started at 0) and makes the results depend on the reactor behavior and MDP policy instead of random starting conditions.
2. Compute P once.
3. For each time step: call **control_iteration** to get the best action, randomly sample the next state using the action's transition probabilities, record the normalized power as the response, and update the current state.

2.4 Evaluation Metrics

Four metrics measure how well the reactor response tracks the demand curve.

Mean Absolute Error (MAE)

$$\text{MAE}(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

Mean Squared Error (MSE)

$$\text{MSE}(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

R² Coefficient

$$R^2(y, \hat{y}) = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

Pearson Correlation Coefficient

$$\text{Corr}(y, \hat{y}) = \frac{\text{Cov}(y, \hat{y})}{\sigma(y) \cdot \sigma(\hat{y})}$$

2.5 Experimentation and Results

All experiments used $\gamma = 0.9$, random seed = 42, and 512 demand samples.

Table 2: Summary of results across reactors R0-R3

Reactor	Model	MAE	MSE	R ²	Pearson
R0	RBMK	0.0261	0.0015	0.9862	0.9952
R1	RBMK-II	0.0077	0.0002	0.9986	0.9993
R2	RBMK-PROTO	0.0150	0.0005	0.9952	0.9979
R3	RBMK-BROKEN	0.3906	0.2317	-1.1732	0.3254

2.5.1 R0 (RBMK)

Probabilities: decrease = [0.55, 0.20, 0.25] | maintain = [0.95, 0.025, 0.025] | increase = [0.65, 0.25, 0.10]

According to Figure 3, this reactor tends to produce lower power levels in response to the demand. Figure 4 shows moderate scatter around the regression line. Figure 5 shows that the actions don't typically follow their desired action. The emphasis in the graph is that maintain preforms the worst, as it frequently results in unintended state changes rather than keeping the reactor stable. These graphs and the MAE value of 0.0261 and the R^2 of 0.9862 show that the reactor is able to adapt to the changed environment moderately well.

2.5.2 R1 (RBMK-II)

Probabilities: decrease = [0.10, 0.60, 0.30] | maintain = [0.00, 0.95, 0.05] | increase = [0.10, 0.80, 0.10]

All three actions work reliably: maintain succeeds 95% of the time, increase succeeds 80% of the time, and decrease succeeds 60% of the time.

As shown in Figure 6, the reactor response closely follows the demand curve throughout the simulation with very deviation. Figure 7 shows an extremely tight correlation around the regression line (with a R^2 of 0.9986), indicating highly accurate tracking behavior. Figure 8 also demonstrates balanced and near-ideal stochastic dynamics across all three actions, especially the strong maintain and increase probabilities. Together, the graphs show that R1 is very reliable and consistently meeting the changing power demand.

2.5.3 R2 (RBMK-PROTO)

Probabilities: decrease = [0.175, 0.025, 0.800] | maintain = [0.050, 0.600, 0.350] | increase = [0.000, 0.200, 0.800]

As shown in Figure 9 the graph tracks the power demand well, except at times when there is a large drop in power demand in which the reactor can't keep up with the rate of decrease. Figure 10 shows a strong linear relationship between demand and response (with an R^2 of 0.9952), indicating that the reactor still maintains accurate tracking despite its stochastic behavior. Figure 11 clearly shows that the maintain action is the most stable whereas increase and decrease less often perform their desired actions. Overall, this reactor is good at increasing and maintaining but cannot track large decreases.

2.5.4 R3 (RBMK-BROKEN)

Probabilities: decrease = [0.00, 0.01, 0.99] | maintain = [0.00, 1.00, 0.00] | increase = [0.99, 0.01, 0.00]

Decrease and increase almost always do nothing in this reactor, with both actions remaining unchanged 99% of the time. As a result, the maintain action is the only reliable option, so the reactor tends to stay near the same power level instead of adjusting to demand. Figure 12 shows this clearly, as the response curve stays nearly flat while the demand curve changes over time. In Figure 13, the scatter plot forms mostly horizontal clusters instead of a diagonal trend. Figure 14 also reflects this behavior, since the radar chart is dominated by the maintain axis while decrease and increase contribute almost nothing. This leads to poor overall performance as also shown by the very low R^2 of -1.1732 which is worse than guessing.

3. Extra Edge Cases

3.1 Experimentation and Results

Reactor	MAE	MSE	R ²	Pearson
R4	0.1728	0.0447	0.5805	0.9156
R5	0.0115	0.0003	0.9974	0.9987
R6	0.0497	0.0057	0.9464	0.9837
R7	0.0164	0.0005	0.9953	0.9977
R8	0.0055	0.0000	0.9996	0.9998

Table 3: Summary of results across edge case reactors

3.2 R4

Probabilities: decrease = [0.0, 0.2, 0.8] | maintain = [0.0, 1.0, 0.0] | increase = [0.8, 0.2, 0.0]

We tested this edge case to test what happens when the MDP almost always maintains. All three actions tend to maintain most of the time. This means that the MDP cannot depend on an increase to increase or decrease.

Figure 15 shows that the reactor is able to follow the direction of the demand curve, but the response changes with failure to meet the magnitude because all three actions tend to maintain the current state most of the time (as seen by a MAE value of 0.1728). Instead of making large adjustments, the MDP can only gradually increase or decrease through repeated small changes. Figure 16 reflects this, as the scatter still follows a positive trend but forms curved step-like clusters rather than a tight diagonal line. Figure 17 shows why this occurs, since the maintain probabilities dominate the radar chart, meaning the reactor is much more likely to stay at the same level than successfully increase or decrease.

3.2.1 R5

Probabilities: decrease = [0.333, 0.334, 0.333] | maintain = [0.333, 0.334, 0.333] | increase = [0.333, 0.334, 0.333]

We tested this edge case to test what happens when the MDP has almost no real control over the system. All three actions have the same chance of causing any outcome, so the MDP cannot depend on any action to reliably increase, decrease, or maintain the reactor power.

Figure 18 shows that the reactor is still able to follow the overall demand trend even though every action behaves randomly. Since no action reliably increases, decreases, or maintains the reactor power, the MDP adapts by relying on long-term averaging effects rather than precise control at each timestep. Figure 19 reflects this with a strong overall linear correlation, although the scatter is slightly more spread out due to the randomness in the transitions. Figure 20 shows balanced behavior across all three actions because each action has identical probabilities, meaning the reactor has no reliable control action.

3.2.2 R6

Probabilities: decrease = [0.00, 0.00, 1.00] | maintain = [0.00, 1.00, 0.00] | increase = [0.00, 1.00, 0.00]

We tested this edge case to test what happens when one of the actions behaves incorrectly. In this case, the decrease action actually never decreases the power level. This means that the reactor has no real way to reduce power through the decrease or the maintain action.

Figure 21 shows that the reactor can increase power effectively, but when the reactor needs to decrease it is unable to. Since the decrease action never actually decreases the reactor power, the MDP adapts by relying almost entirely on increase and maintain actions to follow demand as closely as possible. Figure 22 still shows a strong positive relationship between demand and response, although the flat clustering near the bottom and top reflects the reactor's inability to decrease. Figure 23 highlights this imbalance clearly, with the radar chart showing that only the maintain and increase actions behave correctly while the decrease action does not.

3.2.3 R7

Probabilities: decrease = [0.50, 0.00, 0.50] | maintain = [0.50, 0.00, 0.50] | increase = [0.50, 0.00, 0.50]

We tested this edge case to examine what happens when the reactor can never do its desired action. For example, if it wants to increase, it can only decrease or maintain.

Figure 24 shows that the reactor is still able to roughly follow the overall demand trend, but the response fluctuates constantly because none of the actions perform their intended behavior. A decrease action can only maintain or increase, a maintain action can only increase or decrease, and an increase action can only decrease or maintain, so the MDP cannot make direct controlled adjustments. Instead, the MDP adapts by taking advantage of the balanced randomness in the reactor, where the constant increases and decreases eventually average out over time and keep the reactor close to the target demand. Figure 25 reflects this with a generally linear trend but noticeably wider scatter caused by the continuous unintended state changes, while Figure 26 highlights the absence of any reliable action behavior across all three control actions.

3.2.4 R8

Probabilities: decrease = [0.95, 0.05, 0.00] | maintain = [0.10, 0.80, 0.10] | increase = [0.00, 0.05, 0.95]

We tested this edge case to see what happens when the MDP almost always overshoots its desired action. For example, when the desired action is increase, 95% the power level will increase by 2.

Figure 27 shows that the reactor tracks the demand extremely well despite the large overshoots in the increase and decrease actions. Since the reactor almost always changes by two levels instead of one, the MDP adapts by carefully timing when to increase, decrease, or maintain so the larger jumps still keep the response near the target demand. Figure 28 reflects this with an almost perfect linear relationship between demand and response, showing that the MDP successfully compensates for the reactor behavior. Figure

29 shows that the maintain action is the most stable behavior, while the increase and decrease axes remain small because those actions usually overshoot by two levels instead of performing the desired one-level adjustment.

4. Conclusions

The experimentation carried out across the nine reactors provided allows for a comprehensive evaluation of the MDP based control system under a wide variety of stochastic conditions. The results obtained, summarized through the four regression quality metrics (MAE, MSE, R^2 and Pearson correlation coefficient), reveal a clear relationship between the stochastic dynamics of a reactor and the ability of the control system to track the power demand curve.

In general, the control system performs well when the transition probabilities associated with each action are consistent and predictable, even in cases where the reactor does not behave deterministically. An example of this is reactor R8, which achieves the best results across all four metrics (MAE = 0.0055, $R^2 = 0.9996$) despite having a strong tendency to overshoot by two levels when executing the increase and decrease actions. The Value Iteration algorithm, having full knowledge of the transition probability matrix P , is capable of anticipating this systematic overshoot and adjusting its policy accordingly, choosing actions at states that account for the expected displacement.

It is therefore concluded that the performance of the MDP based control system is not determined solely by the quality of the policy computed, but is equally dependent on the stochastic characteristics of the reactor being controlled. A reactor whose actions are reliable, directionally consistent, and can go in every direction, even if imprecise in magnitude, can be controlled effectively.

Conversely, reactors where only one action behaves reliably, improving the control algorithm alone is insufficient, as the reactor transition probabilities themselves limit the MDP's ability to effectively regulate the system. This behavior is demonstrated by reactors such as R3, where the decrease and increase actions almost always fail to change the power level, leaving the reactor largely unable to respond to changes in demand. As shown in Figure 12, the response curve remains nearly flat while the demand varies significantly over time, indicating that the MDP cannot meaningfully adjust the reactor state despite computing the optimal policy. In these situations, improving the control algorithm alone is insufficient, and the reactor dynamics themselves would need to be corrected before reliable tracking performance could be achieved.

5. Use of AI

Our group used AI to confirm our understanding of the evaluation metrics and what they demonstrate. This helped us make sure we were basing our conclusions on the correct foundation and not a misinterpreted ones. In addition, we also used AI to better understand specific details such as the role of the discount factor $= 0.9$ and the purpose of random seed values. To create our report we used LaTeX formatting and used AI to learn the syntax of it.

All information obtained through AI was cross-checked against reliable sources.

6. Appendix

6.1 R0(RBMK)

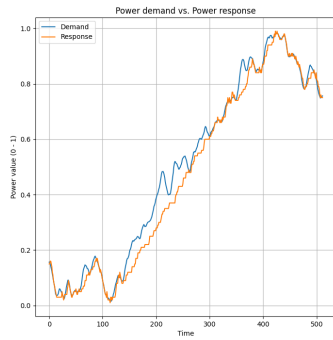


Figure 3

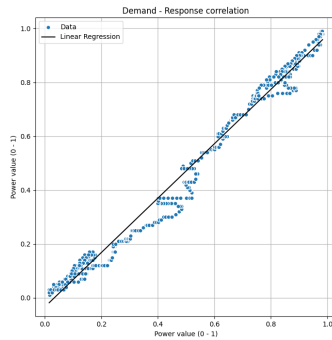


Figure 4

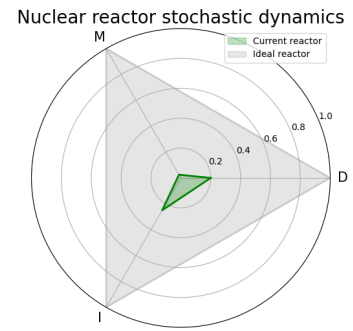


Figure 5

6.2 R1(RBMK-11)

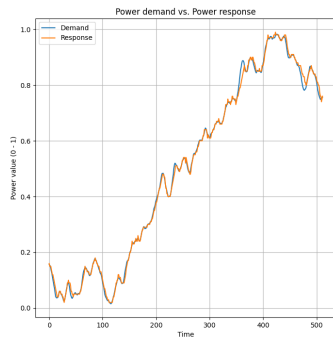


Figure 6

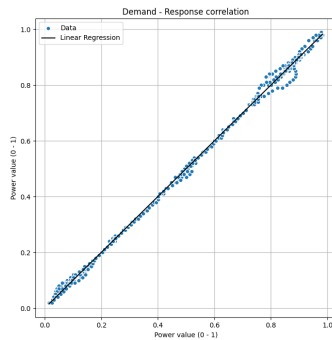


Figure 7

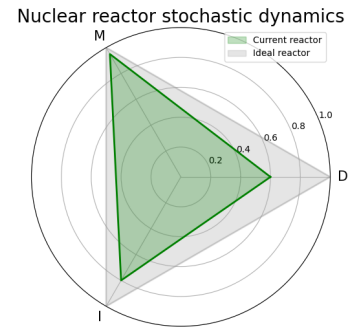


Figure 8

6.3 R2(RBMK-PROTO)

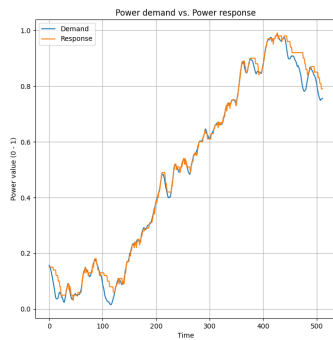


Figure 9

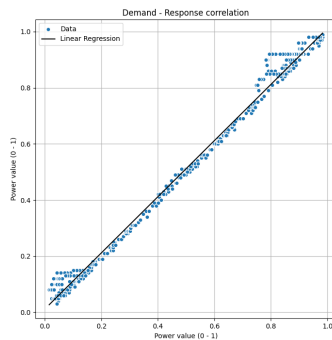


Figure 10

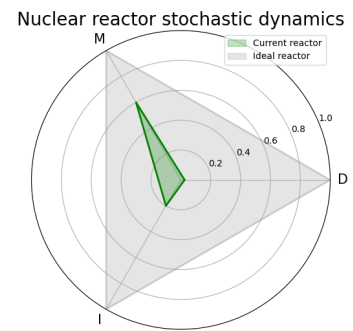


Figure 11

6.4 R3(RBMK-BROKEN)

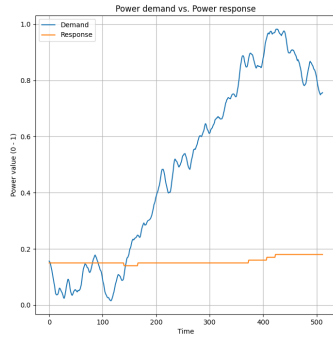


Figure 12

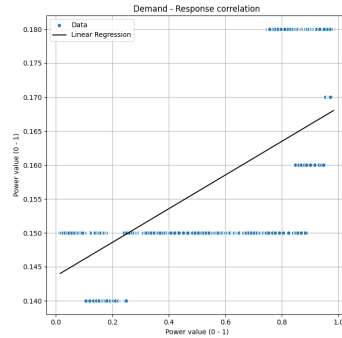


Figure 13

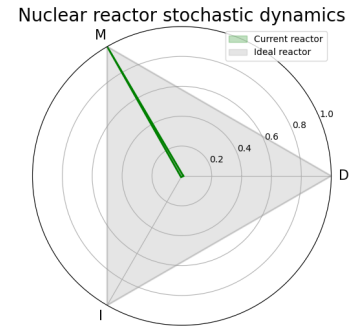


Figure 14

6.5 R4

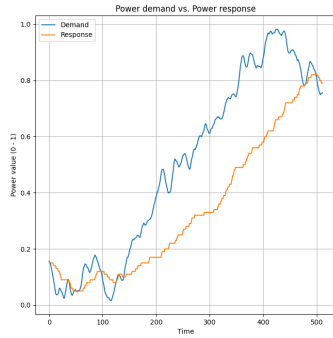


Figure 15

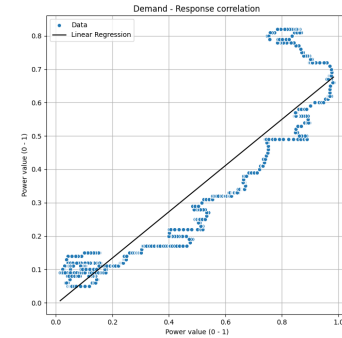


Figure 16

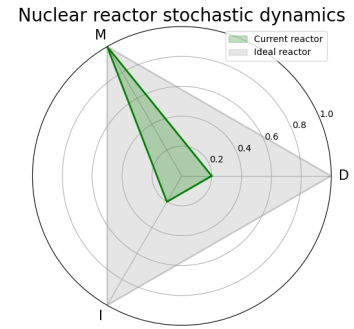


Figure 17

6.6 R5

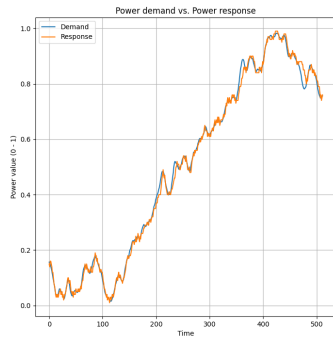


Figure 18

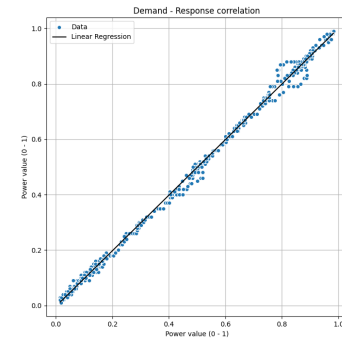


Figure 19

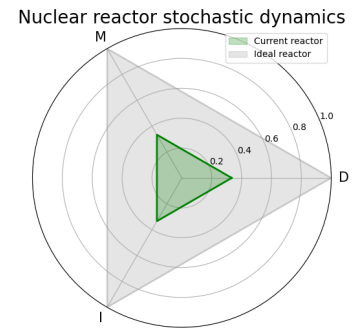


Figure 20

6.7 R6

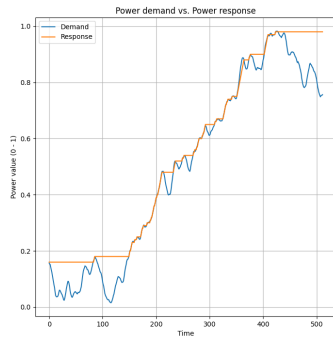


Figure 21

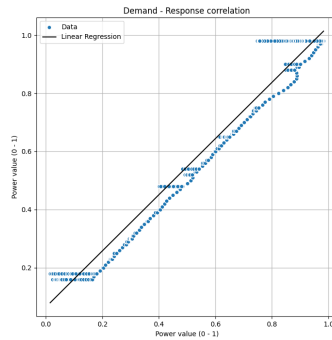


Figure 22

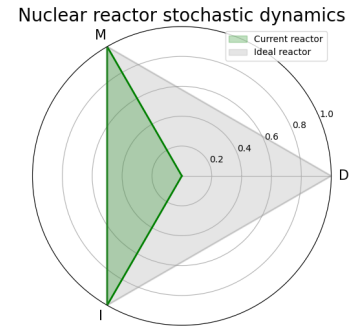


Figure 23

6.8 R7

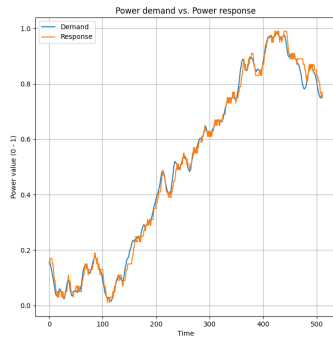


Figure 24

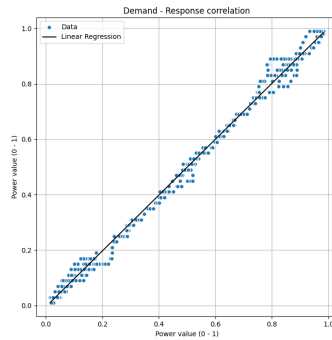


Figure 25

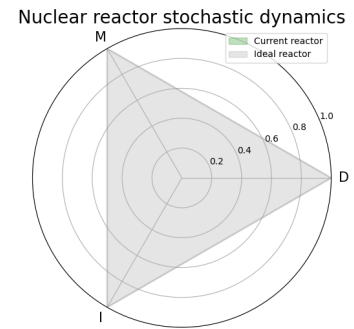


Figure 26

6.9 R8

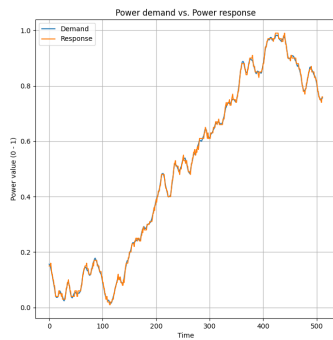


Figure 27

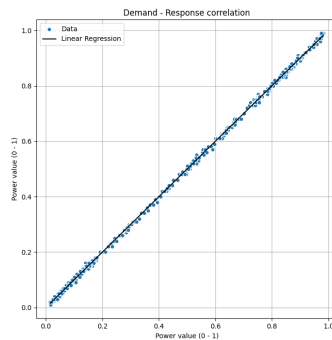


Figure 28

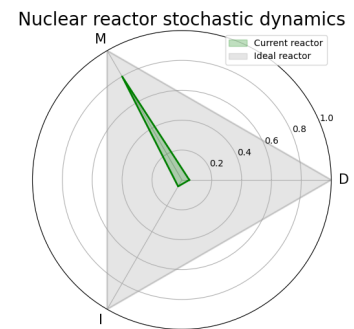


Figure 29